256 hw

2025-04-17

```
setwd("C:/Users/Cynthiali0621/Desktop")
```

```
#question1
if(!require("Matrix")) {install.packages("Matrix"); library("Matrix")}
```

```
## Loading required package: Matrix
```

```
if(!require("lme4")) {install.packages("lme4"); library("lme4")}
```

```
## Loading required package: lme4
```

```
data <- read.csv("gababies.csv")
model <- glmer(lowbrth ~ initage + timesnc + birthord + (1 | momid),
               data = data,
               family = binomial(link = "logit"))
summary(model)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: lowbrth ~ initage + timesnc + birthord + (1 | momid)
##    Data: data
##
##      AIC      BIC   logLik deviance df.resid
##   1193.0   1217.6   -591.5   1183.0      995
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.6992 -0.5782 -0.3415  0.6792  2.6610
##
## Random effects:
##  Groups Name        Variance Std.Dev.
##  momid  (Intercept) 2.311    1.52
## Number of obs: 1000, groups:  momid, 200
##
## Fixed effects:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  1.70425    0.80660   2.113  0.03461 *
## initage     -0.12500    0.04445  -2.812  0.00492 **
## timesnc     -0.03025    0.03644  -0.830  0.40649
## birthord    -0.05800    0.09132  -0.635  0.52533
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr) initag timsnc
## initage  -0.959
## timesnc   0.111 -0.023
## birthord -0.225  0.031 -0.787
```

```
#question2
newdata <- data.frame(initage = 15, timesnc = 10, birthord = 5)
# Predict log-odds (type="link") and probability (type="response") without random effects
log_odds <- predict(model, newdata = newdata, re.form = NA, type = "link")
prob <- predict(model, newdata = newdata, re.form = NA, type = "response")

log_odds
```

```
##          1
## -0.7632046
```

```
prob
```

```
##         1
## 0.3179509
```

For a child with these characteristics, the estimated probability of having low birthweight is about 31.8%, based only on the fixed effects.

```
#question3
between_var <- 2.311
within_var <- pi^2 / 3

icc <- between_var / (between_var + within_var)
icc
```

```
## [1] 0.4126146
```

The ICC of 0.4126146 means that 41.3% of the total variance in the odds of low birthweight is due to differences between mothers.